

A Practical Guide to Denoising diffusion probabilistic model

TOPIC -- DENOISING DIFFUSION PROBABILISTIC MODEL

-- 01 --

To construct a probability path, we start by construct a conditional probability path $p_t(x|z)$ and the corresponding conditional velocity field $v_t(x|z)$ on some conditional distribution $q(z)$. A natural choice is the Gaussian conditional probability path:

-- 02 --

Samplers Given a diffusion model, one may regard it either as a continuous process, and sample from it by integrating a SDE, or one can regard it as a discrete process, and sample from it by iterating the discrete steps. The choice of the "noise schedule" β_t can also affect the quality of samples. A noise schedule is a function that sends a natural number to a noise level: $t \mapsto \beta_t, t \in \{1, 2, \dots\}, \beta \in (0, 1)$. A noise schedule is more often specified by a map $t \mapsto \sigma_t$. The two definitions are equivalent, since $\beta_t = 1 - \frac{\sigma_t^2}{\sigma_{t-1}^2}$. In the DDPM perspective, one can use the DDPM itself (with noise), or DDIM (with adjustable amount of noise). The case where one adds noise is sometimes called ancestral sampling. One can interpolate between noise and no noise. The amount of noise is denoted η ("eta value") in the DDIM paper, with $\eta = 0$ denoting no noise (as in deterministic DDIM), and $\eta = 1$ denoting full noise (as in DDPM). In the perspective of SDE, one can use any of the numerical integration methods, such as Euler–Maruyama method, Heun's method, linear multistep methods, etc. Just as in the discrete case, one can add an adjustable amount of noise during the integration. A survey and comparison of samplers in the context of image generation is in.

-- 03 --

Flow-based diffusion model Abstractly speaking, the idea of diffusion model is to take an unknown probability distribution (the distribution of natural-looking images), then progressively convert it to a known probability distribution (standard Gaussian distribution), by building an absolutely continuous probability path connecting them. The probability path is in fact defined implicitly by the

score function $\nabla \ln p_t$. In denoising diffusion models, the forward process adds noise, and the backward process removes noise. Both the forward and backward processes are SDEs, though the forward process is integrable in closed-form, so it can be done at no computational cost. The backward process is not integrable in closed-form, so it must be integrated step-by-step by standard SDE solvers, which can be very expensive. The probability path in diffusion model is defined through an Itô process and one can retrieve the deterministic process by using the Probability ODE flow formulation. In flow-based diffusion models, the forward process is a deterministic flow along a time-dependent vector field, and the backward process is also a deterministic flow along the same vector field, but going backwards. Both processes are solutions to ODEs. If the vector field is well-behaved, the ODE will also be well-behaved. Given two distributions π_0 and π_1 , a flow-based model is a time-dependent velocity field $v_t(x)$ in $[0, 1] \times \mathbb{R}^d$, such that if we start by sampling a point $x \sim \pi_0$, and let it move according to the velocity field:

-- 04 --

Conditioning The base diffusion model can only generate unconditionally from the whole distribution. For example, a diffusion model learned on ImageNet would generate images that look like a random image from ImageNet. To generate images from just one category, one would need to impose the condition, and then sample from the conditional distribution. Whatever condition one wants to impose, one needs to first convert the conditioning into a vector of floating point numbers, then feed it into the underlying diffusion model neural network. However, one has freedom in choosing how to convert the conditioning into a vector. Stable Diffusion, for example, imposes conditioning in the form of cross-attention mechanism, where the query is an intermediate representation of the image in the U-Net, and both key and value are the conditioning vectors. The conditioning can be selectively applied to only parts of an image, and new kinds of conditionings can be finetuned upon the base model, as used in ControlNet. As a particularly simple example, consider image inpainting. The conditions are $\tilde{x}$, the reference image, and m , the inpainting mask. The conditioning is imposed at each step of the backward diffusion process, by first sampling $x_t \sim N(\alpha_t \tilde{x}, \sigma_t^2 I)$, a noisy version of $\tilde{x}$, then replacing x_t with $(1 - m) \odot x_t + m \odot \tilde{x}$, where $\odot$ means elementwise

multiplication. Another application of cross-attention mechanism is prompt-to-prompt image editing. Conditioning is not limited to just generating images from a specific category, or according to a specific caption (as in text-to-image). For example, demonstrated generating human motion, conditioned on an audio clip of human walking (allowing syncing motion to a soundtrack), or video of human running, or a text description of human motion, etc. For how conditional diffusion models are mathematically formulated, see a methodological summary in.

-- 05 --

$\frac{d}{dt}\varphi_t(x) = v_t(\varphi_t(x)) \quad t \in [0, 1]$, starting from $\varphi_0(x) = x$ $\{\text{displaystyle}$
 $\frac{d}{dt}\varphi_t(x) = v_t(\varphi_t(x)) \quad t \in [0, 1], \quad \text{starting from}$
 $\varphi_0(x) = x\}$